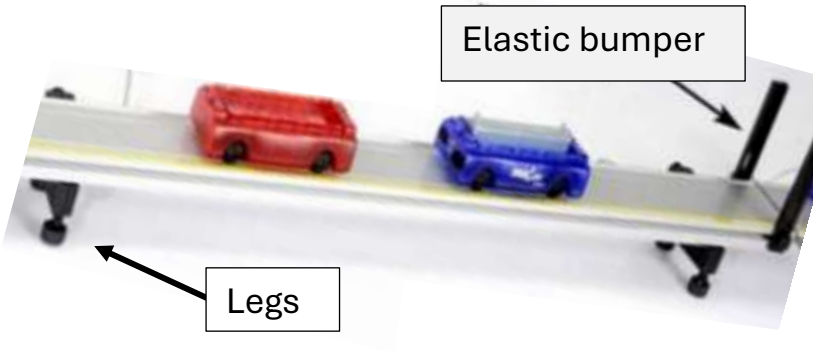
Manual to Lab 3: PHY2048C

Florida State University

1D-Collisions



About labs in this class

The labs in this class will have general instructions, and many things need to be figured out by the students. **Answer the questions and record all your measurements in your lab notebook and then submit the scan of the notebook to Canvas. Also upload screenshots of the *phyphox* plots or the frame of the collision of your video.**

About this lab

In this lab, you will measure the coefficient of restitution of a partially inelastic collision. You are provided with all the tools required to make this measurement. You will need the camera of your phone and/or the app *phyphox*. You will take screenshots of your video or the *phyphox* plots you get.

Activity 1. Set up the collision experiment. Make sure it is perfectly leveled. Perform one perfectly elastic collision experiment with two equal masses. Describe what you see in your notebook. (Hint: the elastic bumpers will make your life easier. Use the legs to level the setup).

Question 1. Is this behavior consistent with the equations of conservation of momentum and conservation of energy? Use the equations to prove this.

Activity 2. Come up with a way to use your phone to measure the speed of the cars before and after the impact *by taking a video or using phyphox* and confirm the equation you used in Question 1 works. (Remember to take the screenshots)

Activity 3. Perform a totally inelastic collision by colliding the sides of the cars that have Velcro patches.

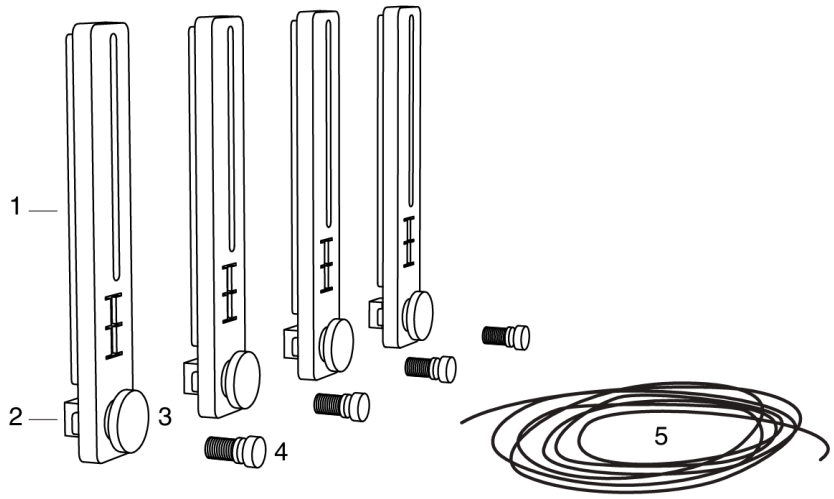
Question 2. Get a percentage discrepancy between prediction and observation for the velocity after the collision of at most 8%.

Activity 3. Now collide the magnet side with the velcro side. This is a partially inelastic collision. Measure the coefficient of restitution of the collision, given by $\epsilon = \sqrt{\frac{KE_f}{KE_i}}$. Make multiple measurements and a histogram (use other group's values as well) and report the average ϵ value with at least an 10% error.

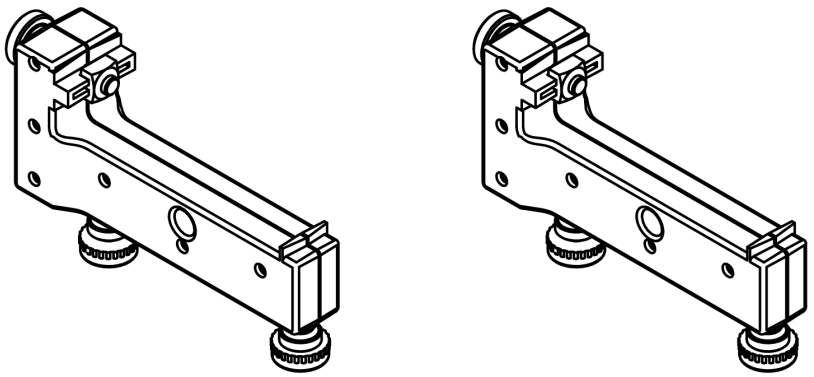
Collisions Lab Piece References

For organizing at the end of the lab

Elastic Bumper



Dynamics Track Feet



Dynamics Track End Stop

